Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/05/2005
PI Name:	Colwell, Joshua Ph.D.		
Project Title:	Dynamics of Charged Dust Near Surfaces in Space		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	josh.colwell@lasp.colorado.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	303-492-6805
Organization Name:	University of Colorado		
PI Address 1:	Laboratory for Atmospheric and Space Physics		
PI Address 2:	392 UCB		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80309-0392	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	01/17/2002	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
Contact Email:	brian.j.motil@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Robertson, Scott (University of Colorado) Horanyi, Mihaly (University of Colorado)		
Grant/Contract No.:	NAG3-2716		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We will perform experimental and numerical studies of the levitation and transport of charged dust above planetary surfaces and spacecraft surfaces in the near-surface plasma environment. Experimental studies will be performed in a plasma device using simulated planetary and spacecraft surfaces with a plasma sheath over the surface. Direct numerical simulations of the particle dynamics will also be performed. We will identify and characterize the conditions under which dust is levitated and redeposited over various realistic planetary and spacecraft surfaces. We will conduct an experimental study of conditions which aid levitation. This includes varying surface properties, regolith density and size distribution, and disturbances to the surface from direct impacts and seismic activity (produced, for example, by large impacts elsewhere on the body). We will quantify the transport of dust across the terminator under various conditions and apply these results to the observed distribution of regolith on planetary surfaces, such as asteroids and planetary satellites. We will simulate conditions for horizontal transport between different terrain types including topography, surface properties, and plasma environments. The presence of charge on dust on planetary surfaces plays a crucial role in two basic processes in the evolution of planetary regoliths: (1) levitation of dust particles from the surface; and (2) transport or escape of particles from the surface. Understanding the conditions which lead to dust levitation is necessary not only for understanding the origin of such diverse processes as spoke formation in SaturnÃ??s rings and the lunar horizon glow, but also for preventing contamination of instrumentation placed on planetary surfaces, such as spacecraft landers and rovers. These two processes are fundamental to a basic understanding of the formation and evolution of the near-surface regolith of moons, asteroids, and ring particles, and to the interpretation of remote sensing data of these objects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have performed numerical simulations of the transport of dust near surfaces in space and applied this to the Moon and asteroids. We have a new three-dimensional simulation that allows simulations of more realistic landforms. We have performed experiments to measure the density profile of photoelectron sheaths and the charging of isolated dust grains in photoelectron sheaths. We have performed experiments on the charging of grains in a regolith through triboelectric charging and in a plasma sheath as well. These results have been published in peer-reviewed journals and presented at numerous scientific meetings. We have developed a new experimental apparatus to study the charging properties of dust on the lunar surface. We have performed experiments on the levitation and horizontal transport of dust in a plasma sheath. These results have been presented at scientific conferences. We have written a review article on the problem of charged dust on and near the lunar surface.
Bibliography Type:	Description: (Last Updated: 12/05/2005)
Articles in Peer-reviewed Journals	Mitchell, C.; Horanyi, M.; Colwell, J. "Tenuous Ring Formation by the Capture of Interplanetary Dust at Saturn" J. Geophys. Res., Vol. 110, No. A9, A09218 , Sep-2005
Articles in Peer-reviewed Journals	Colwell, J. E.; Gulbis, A. A. S.; Horanyi, M.; Robertson, S. "Dust Transport in Photoelectron Layers and the Formation of Dust Ponds on Eros" Icarus, Vol. 175, pp. 159-169. , May-2005
Presentation	Colwell, J. E., Horanyi, M., Robertson, S., Wheeler, P. "Behavior of Charged Dust in Plasma and Photoelectron Sheaths" Conference on Dust in Planetary Systems, Kaua'i, Hawaii, Sep. 26-30, 2005. Sep-2005